Loss function Two classes Hinge loss

$$\begin{cases} 0 & \text{if correct } y_i = \hat{y}_i \\ -y_i \hat{y}_i & \text{if incorrect } y_i \neq \hat{y}_i \end{cases}$$

$y_i \in \{-1, +1\}$

$\hat{y}_i \in \{-1, +1\}$

multi class classification

Cross entropy loss

$$H(X) = -\sum_{x \in X} p(X=x) \log\left(p(X=x)\right)$$

$X = 1, 2, 3, 4, 5, 6$

Roll a dice $p(X) = \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \dots$

$$= -\sum_{x=\{1,2,3,4,5,6\}} \frac{1}{6} \log\left(\frac{1}{6}\right)$$

$$H(X) = -\log\left(\frac{1}{6}\right) = \log(6)$$ / measure of randomness

Roll a dice

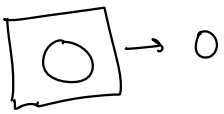
$P(X=1) = 1$

$P(X=2) = 0$

$P(X=3) = 0$

$$H(X) = \sum_{x \in \{1,\dots 6\}} 1 \underbrace{\log 1}_{\downarrow 0} + \checkmark\, 0 \underbrace{\log 0^{+}}_{\downarrow -\infty} + \dots$$

$$H(X) = 0$$



$\rightarrow 0$

$$l(y_i, \hat{y}_i) = H(Y, \hat{Y}) = -\sum_{y \in Y} \underbrace{P(Y=y)}_{\text{True labels are typically deterministic}} \log\left(P(\hat{Y}=y)\right)$$

$$l(y_i, \hat{y}_i) = -\log\left(\underbrace{P(\hat{y}_i = y_i)}_{\text{Predict the probability of label } \hat{y}_i \text{ taking all possible values}}\right)$$

$\hat{y}_i$	$P(\hat{y}_i)$
0	0.1
1	0.2
2	0.3

In multiclass classification the output has to be a vector of probabilities.

$$P(Y=y_i) \geq 0, \quad \sum_{y \in Y} P(Y=y) = 1$$

$$\text{softmax}\left(\underbrace{[z_1, z_2, z_3 \ldots, z_{10}]}_{\underline{z}}\right) = \frac{[\exp(z_1), \exp(z_2), \ldots]}{\sum_{i=1}^{10} \exp(z_i)}$$